

phase_08C_GAP_AUDIT

Phase 8C — Code-Grounded Gap Audit

Honest completion: ~45% of Phase 8C deliverables. Crypto/trust primitives (PQ E2EE envelope, software proof-of-GPU, software TEE attestation) are genuinely implemented with real-behavior tests in the vasic-digital/security submodule; scheduler cost/preemption and a fiber transport exist. But the higher pillars — LLMOrchestrator serving (still a stub), GPU resource model, model-integrity gate, marketplace/audit, inferenceproxy, admission — are MISSING, and the implemented primitives lack the Chutes byte-interop, O(1) spot-check, streaming, and attestation-gated scheduling the roadmap specifies.

Audited 2026-06-01 against actual code in pkg/, internal/, cmd/, and the security/ submodule (vasic-digital/security). A deliverable is **DONE** only if implemented AND covered by real-behavior tests; stub/partial/untested → PARTIAL/MISSING.

3-Axis Package Status (Refreshed 2026-06-04, HXC-939)

Why this section exists: “exists” ≠ “used”. wired is **measured** via `go list -deps ./cmd/... | grep -Fx <module-path>/<pkg>` (module = github.com/HelixDevelopment/helix_cluster), not assumed. **“Completed (registry) ≠ wired”** — Completed only means source+tests exist; it does NOT prove a shipped binary reaches it.

STALENESS CORRECTION: the 2026-06-01 table below marked pkg/modelintegrity (then hf_cache_verify), pkg/inferenceproxy, and pkg/marketplace as MISSING. As of 2026-06-04 **all three EXIST and are TESTED** — but **all three are ORPHANED** (not reachable from any cmd/ binary). The registry would show them “Completed”; the measured wired column shows they are exists-but-unused. The rows below override the older prose.

Package	implemented	wired (reachable from cmd/)	tested
pkg/modelintegrity	yes	NO (orphaned)	yes

Package	implemented	wired (reachable from cmd/)	tested
pkg/inferenceproxy	yes	NO (orphaned)	yes
pkg/marketplace	yes	NO (orphaned)	yes
pkg/fiber (miner[?]validator transport)	yes	NO (orphaned)	yes
pkg/scheduler (cost_gpu + preempt)	yes	yes	yes
pkg/resources (GPU model — no attested/multiplier yet)	yes	yes	yes
internal/llm (router — stub Inference)	yes	yes	yes
security/pkg/e2ee (separate module)	yes	NO (not in helix cmd graph)	yes
security/pkg/gpuattest (separate module)	yes	NO (not in helix cmd graph)	yes
security/pkg/attestation (separate module)	yes	NO (not in helix cmd graph)	yes
security/pkg/admission / signed-image gate	NO (absent)	n/a	n/a

Orphan callouts: pkg/modelintegrity, pkg/inferenceproxy, pkg/marketplace, pkg/fiber are implemented+tested but reachable from no binary. They satisfy “the package exists” and would pass go test, but deliver zero end-user value until a binary (e.g. internal/llm’s LoadModel, or the gateway) imports them — exactly the gap the 3-axis model exposes that “Completed” hid.

Deliverable Status Table

Deliverable (roadmap §)	Status	Evidence (file:line)	Notes
8C.1 pkg/security/e2ee PQ envelope (ML-KEM-768 + HKDF-SHA256 + AEAD)	PARTIAL	security/pkg/e2ee/package.go:103–326; tests security/pkg/e2ee/package_test.go:37–360 (14 tests: tamper/nonce-reuse/AAD/wrong-key rejection)	Real ha + repla vs spec AES-256 §4.1 de Poly1305 no gzip

Deliverable (roadmap §)	Status	Evidence (file:line)	Notes
8C.1 E2EE framed transport	DONE	security/pkg/e2ee/ transport.go:31–77; package_test.go:286–336	discovered NO Ch vectors “byte-f referen Length framing io.Rea guard, r concurr minima Honest HMAC respons replay+ and exp docume (no GP no mat O(1) sp device- (\$4.1). oversub GPU — secret p Real ed docume with no clock-s measur evaluati “softwa hardwa (Quote real TD no attes measur Roadm exit gat real har DEFER
8C.2 pkg/gpuattest software proof-of-GPU (challenge/response + fingerprint)	PARTIAL	security/pkg/gpuattest/ package.go:104–323; tests package_test.go:42–380 (13 tests: expired/replay/tamper/ wrong-key/forged-descriptor rejection)	
8C.3 pkg/security/attestation TEE (TDX/NVIDIA quote, measured boot, key release)	PARTIAL	security/pkg/attestation/ attestation.go:1–346; tests attestation_test.go (18 tests)	
8C.3 pkg/security/admission (OPA-style policy + signed-image/cosign)	MISSING	security/pkg/policy/ policy.go is generic rule eval — no admission, no cosign, no signed-image verify; no admission dir anywhere	Only a framew worklo image- absent.
8C.4 pkg/scheduler cost-aware placement	DONE	pkg/scheduler/ cost_gpu.go:28–130; cost_gpu_test.go	Cost-av plugin labels + implem interfac

Deliverable (roadmap §)	Status	Evidence (file:line)	Notes
8C.4 Scheduler preemption + optimistic claiming	PARTIAL	pkg/scheduler/ gang_preempt.go:66–279 (ScheduleGangOptimistic, SchedulePreemptOptimistic, version-conflict)	Priority + optim version tested. is prior roadma multipl SKIP L claimin gated a predica integrat schedul guate
8C.5 pkg/resources GPU type (fingerprint + compute-multiplier catalog + attested/thermal state)	MISSING	pkg/resources/ types.go:23–28 GPUInfo{Count,Model,Memory} only	No dev SUPPORT multipl verified thermal intern has stat detectio attested
8C.6 LLMOrchestrator WRAP vLLM/ SGLang + ranking/failover + thermal pre- warm + trace audit	MISSING	internal/llm/ manager.go:142–158 — Inference returns "[stub inference from %s]"	Honest No vLL backen passthr no failo no trace largest
8C.7 Model-integrity gate (hf_cache_verify → Go, SHA/size)	MISSING	no match for hf_cache/ ModelIntegrity/VerifyModel in pkg/,internal/,security/ pkg/. security/pkg/content/ is text content-filtering, not model integrity	Roadm high-va absent.
8C.7 pkg/marketplace + /audit (registration→bounty→metering→payout, commit-then-prove)	MISSING	no match for bounty/payout/ metering/Marketplace in pkg,internal,cmd	Econon absent.
pkg/inferenceproxy (correlation-ID, trace unwrap, anonymization)	MISSING	no match for correlation/ anonymiz/inferenceproxy in pkg/internal	Absent
pkg/fiber miner [?] validator transport	PARTIAL	pkg/fiber/fiber.go:1–35+; fiber_test.go	Clean-r prefixe with Pi + 1-rea contrac signed gated a

Deliverable (roadmap §)	Status	Evidence (file:line)	Notes
Node identity: verify-then-pin (TOFU) TLS + header sanitization	MISSING	no pin/TOFU/VerifyThenPin in pkg/security/tls.go or security/pkg/security/	underla docume Existin {tls,s cover s SPIFFE keypair then-pi

Top Implementable Gaps (Go, no new infra)

1. **Model-integrity gate** → **pkg/modelintegrity/** (roadmap P1 #7, 0.5wk, highest value/effort ratio) Port hf_cache_verify semantics: given a model directory and a signed manifest of {relpath → sha256, size}, walk the cache and verify every file's SHA-256 and byte size before a model is marked servable; reject on any mismatch/missing/extra file. Pure stdlib (crypto/sha256, io, os). Wire into internal/llm/manager.go LoadModel as a precondition. **Use only V2 HMAC-SHA256 semantics — never MD5 (risk §7 V1).** *Mutation-pairable acceptance test:* build a temp model dir + manifest; assert Verify passes; then (a) flip one byte → expect ErrHashMismatch, (b) truncate a file → expect ErrSizeMismatch, (c) delete a file → expect ErrMissing. Mutation: invert the subtle.ConstantTimeCompare/size check and confirm a test fails.
2. **Attestation-gated scheduler admission predicate** → **pkg/scheduler/** (extend) (roadmap §5.1, P1) Add an AdmissionPredicate hook on Scheduler (and a Plugin.Filter participant) that rejects a node unless it carries a verified attestation token. Define an interface NodeTrust interface{ Verified(nodeID string) bool } satisfied by gpuattest.Verifier / attestation.Verifier results; nodes failing verification are filtered out before scoring. No new infra — operates on existing Node labels + an injected verifier. *Mutation-pairable test:* two nodes, one with a valid gpuattest response registered, one spoofed; assert only the attested node is placeable. Mutation: make the predicate always-return-true and confirm the spoofed-node-rejected test fails.
3. **GPU resource model** → **pkg/resources/** (extend GPUInfo) (roadmap 8C.5, P1) Extend GPUInfo with DeviceFingerprint string, ComputeMultiplier float64 (from a static SUPPORTED_GPUS catalog keyed by Model), Attested bool, and Thermal{TempC int; Ready bool}. Provide LookupMultiplier(model string) against a hardcoded catalog (H100/H200/A100/etc.) with a conservative default. Feed ComputeMultiplier/Attested into the cost scorer (#2 and cost_gpu.go). *Mutation-pairable test:* assert LookupMultiplier("H100") > LookupMultiplier("A100") > default; assert an un-attested GPU is scored below an attested one. Mutation: swap two catalog entries → ordering test fails.
4. **E2EE Chutes byte-interop vectors + ChaCha20-Poly1305 cipher option** → **security/pkg/e2ee/** (roadmap §8 exit gate, P0) Add a testdata/ corpus of fixed (encap-key, KEM-ct, nonce, AAD, plaintext, expected-ciphertext) vectors and a vector-driven test asserting byte-for-byte Seal/Open reproduction; add a

ChaCha20-Poly1305 AEAD variant (golang.org/x/crypto/chacha20poly1305) selectable via SessionConfig, matching the roadmap's stated cipher. Closes the "byte-for-byte vs reference vectors" exit gate without a live worker. *Mutation-pairable test*: table of golden vectors; assert exact ciphertext bytes. Mutation: perturb one HKDF info-label/salt byte in derivation → all golden-vector tests fail.

5. **gpuattest O(1) spot-check + device-sealed encrypt → security/pkg/gpuattest/** (roadmap 8C.2, P0) Add SpotCheck(resp, index) that re-derives and compares a single deterministic challenge-indexed word of the proof (O(1) verification of a large software-PoVW response) and a SealForDevice(deviceKey, plaintext)/OpenFromDevice pair binding an AEAD key to the device secret. Stays software-only and clearly scoped (no GPU kernel). *Mutation-pairable test*: genuine response passes spot-check at random indices; a response with one tampered word fails at that index. Mutation: make SpotCheck ignore index → tampered-word test fails.
6. **fiber signed-identity + stake-gated admission → pkg/fiber/ (extend)** (roadmap §9 Phase 6 hook) Add an ed25519 handshake frame type carrying {nodeID, pubkey, signed-nonce} and an Admit(func(nodeID, stake) bool) gate run before application frames flow; reject unsigned/under-staked peers (stake supplied by caller, no chain). Reuses existing frame machinery. *Mutation-pairable test*: peer with valid signature + sufficient stake admitted; bad signature or zero stake rejected. Mutation: skip signature verification → bad-signature test fails.
7. **inferenceproxy correlation/anonymization → pkg/inferenceproxy/** (roadmap P2 #10) An http.Handler middleware that injects/propagates a correlation-ID header, records which backend served a request (audit trail), and deterministically anonymizes configured request headers/fields (keyed hash, stable per-tenant). Pure stdlib. *Mutation-pairable test*: same input header → same anonymized token across calls; different tenant → different token; correlation-ID present on response. Mutation: make anonymizer return input unchanged → "anonymized ≠ original" test fails.

DEFERRED (need external infra / hardware / GPU kernels / non-Go)

- **8C.3 real TEE attestation (Intel TDX + NVIDIA confidential-GPU quotes, measured boot, attested key release).** Needs real TDX + H100/H200-class confidential-GPU hardware and vendor CA chains (go-tdx-guest/DCAP/NRAS). Software QuoteSigner seam exists; genuine quotes cannot be produced or end-to-end validated in CI without the hardware (roadmap §7, §8 hardware gate).
- **8C.2 hardware-rooted proof-of-GPU (real GraVal CUDA/OpenCL matmul kernel).** Requires GPU + CUDA/OpenCL kernel code (not Go) and a GPU CI runner to prove a spoofed/oversubscribed GPU is rejected (roadmap §8 GPU gate). The software spot-check (#5) is the CPU-CI-testable subset; full hardware validation is deferred.
- **8C.6 vLLM/SGLang runtime WRAP (live serving exit gate).** The Go ranking/failover/passthrough orchestration (gap #6-adjacent) is implementable, but the §8 "live confidential instance / real worker" exit gate requires running

upstream vLLM/SGLang GPU containers — deferred to a GPU integration environment.

- **8C.7 marketplace settlement over real economic flow.** The Go data model + commit-then-prove audit logic is buildable on existing Postgres/etcd, but end-to-end payout/reward reconciliation as an exit gate depends on the (still-missing) serving + metering data plane above; defer the e2e settlement proof until #1/#3/#6 land.